Photon A and the entangled Photon B. For instance, quantum theory allows me to measure whether Photons A and B have the same spin about a vertical axis, without measuring their spins individually. Similarly, quantum theory allows me to measure whether Photons A and B have the same spin about a horizontal axis, without measuring their spins individually. With such a joint measurement, I do not learn Photon A's spin, but I do learn how Photon A's spin is related to Photon B's. And that's important information.

The distant Photon C is entangled with Photon B, so if I know how Photon A is related to Photon B, I can deduce how Photon A is related to Photon C. If I now phone this information to Nicholas, communicating how Photon A is spinning relative to his Photon C, he can determine how Photon C must be manipulated so that its quantum state will match Photon A's. Once he carries out the necessary manipulation, the quantum state of the photon in his possession will be identical to that of Photon A, and that's all we need to declare that Photon A has been successfully teleported. In the simplest case, for example, should my measurement reveal that Photon B's spin is identical to Photon A's, we would conclude that Photon C's spin is also identical to Photon A's, and without further ado, the teleportation would be complete. Photon C would be in the same quantum state as Photon A, as desired.

Well, almost. That's the rough idea, but to explain quantum teleportation in manageable steps, I've so far left out an absolutely crucial element of the story, one I'll now fill in. When I carry out the joint measurement on Photons A and B, I do indeed learn how the spin of Photon A is related to that of Photon B. But, as with all observations, the measurement itself affects the photons. Therefore, I do *not* learn how Photon A's spin was related to Photon B's before the measurement. Instead, I learn how they are related after they've both been disrupted by the act of measurement. So, at first sight, we seem to face the same quantum obstacle to replicating Photon A that I described at the outset: the unavoidable disruption caused by the measurement process. That's where Photon C comes to the rescue. Because Photons B and C are entangled, the disruption I cause to Photon B in New York will also be reflected in the state of Photon C in London. That is the wondrous nature of quantum entanglement, as elaborated in Chapter 4. In fact, Bennett and his collaborators showed mathematically that through its